NewUU Campus Updates

Mobile Programming (Flutter), midterm project proposal. Team BlueBerry.

1. Problem

NewUU students miss class updates and can't find free study rooms because campus information is scattered across three different places: Telegram channels, Hero Study and the EduPage timetable.

A student with five or six courses follows a Telegram channel per course, where room changes and cancellations get buried or muted, and deadlines are announced once. Professors have no simple way to post to exactly one course. Students end up in the wrong room, at cancelled classes, or past deadlines. Separately, the library is full at busy hours, and finding an empty classroom means reading the EduPage timetable room by room.

Why it is local. The problem is specific to NewUU's own mix of tools: Telegram as the de facto announcement channel in Uzbekistan, the university's Hero Study app, and its EduPage timetable.

Target users

- **Students (primary):** NewUU undergraduates taking 4 to 6 courses. They need one place for today's classes, room changes, cancellations and deadlines, plus a free room to study in.
- **Professors (secondary):** NewUU teaching staff running 1 to 4 courses. They need a fast way to post an update to one course and know students saw it.

Evidence

- **Telegram, Oct 13.** A lecture postponement posted at 06:17 had to be forwarded at 07:47. At 08:36 a student replied "Couldn't you have written earlier?"
- **Hero Study** cannot mark a room change or cancellation, and course content cannot be opened from it.
- **The library** is nearly full during the day, which pushes students to look for empty classrooms.

- **Existing tools fail:** Telegram buries updates, Hero Study is not used by professors, EduPage shows schedules rather than answers, and Google Classroom and Moodle are not used for this at NewUU.

2. App concept and user flow

Concept. Professors post short course updates (room change, cancellation, deadline, note) and students see them in one feed, filtered to their own courses. The single most important thing a student can do is open the app and see, on one screen, what changed today. The app also shows which classrooms are free right now, based on the EduPage timetable. The midterm version uses sample data.

Main user flow (student: see what changed today)

1. Open the app and sign in on **Login**, choosing the Student role.
2. Land on **Today feed**: today's classes and every new update for the student's courses, newest first.
3. Tap an update card to expand it and read the details (new room, new time, or note).
4. Tap the course name to open **Course detail** and see that course's earlier updates and deadlines.

Professor flow. Sign in as Professor, tap + on the Today feed to open **Post update**, pick the course and type, fill the fields, and tap Post. The form validates and the update appears at the top of the feed, in under 30 seconds.

Supporting tabs. Deadlines lists every upcoming deadline; **Free rooms** lists free rooms, and a tap opens **Room schedule**.

3. Screens and design

Seven screens cover the flows above (our team has five members, so the rubric asks for one extra screen). Each one is a step in at least one flow.

Figma design: the full mobile UI for all seven screens is in our Figma project.

Screen	Purpose
1. Login	Sign in and pick the role, so the right home opens
2. Today feed	Today's classes and new updates on one screen

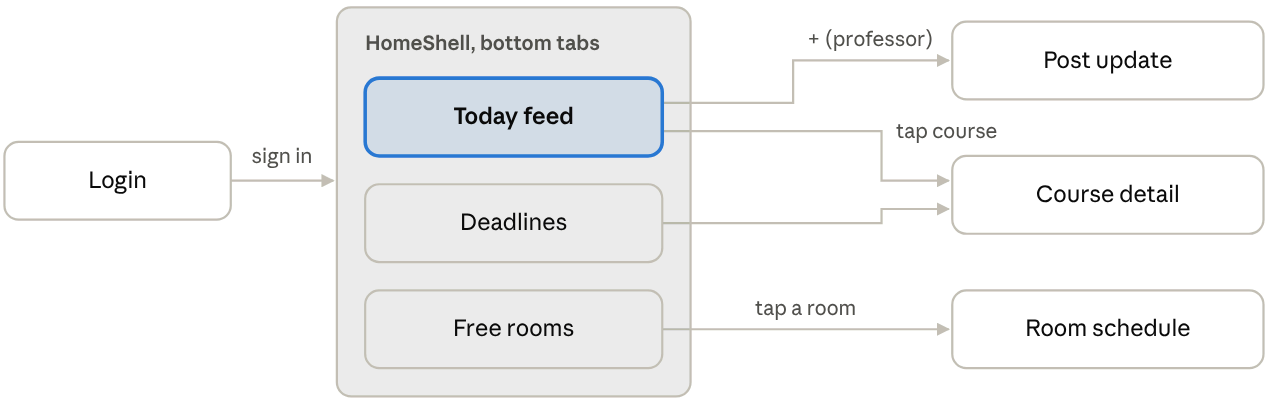
Screen	Purpose
3. Course detail	One course's update history and deadlines
4. Deadlines	Every upcoming deadline, soonest first
5. Post update	A professor posts a typed update in 30 seconds
6. Free rooms	Classrooms free right now, and for how long
7. Room schedule	One room's day, to see how long it stays free

4. Flutter plan

Shared data lives in Riverpod providers; anything only one screen uses stays in that screen's `State` with `setState`. Navigation uses named routes and one bottom navigation bar. Login and Post update are the two forms.

Every screen is one tap from the Today feed

Navigation map: boxes are screens, arrows are Navigator calls



navigation map · 7 screens

Login uses `pushReplacementNamed` so Back does not return to it. The three tabs switch inside `HomeShell`; the other screens open with `pushNamed`, and Post update returns the new update with `Navigator.pop(context, update)`.

State with Riverpod. `ProviderScope` wraps the app. `authProvider` and `updatesProvider` are `NotifierProvider`s; `deadlinesProvider` and `freeRoomsProvider` derive from them with `ref.watch`. Three screens show the same

updates, so the list is lifted into `updatesProvider` and every screen watching it rebuilds when `add()` changes `state`. Screens are `ConsumerWidget`s: `ref.watch` in `build`, `ref.read(...notifier)` in callbacks.

Screen	Main widgets	Local state	Form and validation
Login	`Column`, `SegmentedButton`, 2 `TextFormField`s	Role	`GlobalKey<FormState>`; email ends in `@newuu.uz`; password at least 6 characters
Today feed	`ListView.builder`, `ExpansionTile`, FAB (professor)	None	None
Course detail	`TabBar`, `TabBarView`, `ListView.builder`	None	None
Deadlines	`FilterChip` row, `ListView.builder`	Course filter	None
Post update	`DropdownButtonFormField`, `ChoiceChip`, date picker	Type, date	Course required; room like `B-204`; date not past; message max 200
Free rooms	`ChoiceChip`, `GridView.builder`	Filters	None
Room schedule	`ListView.builder` of `Row`s, `Stack`	None	None

5. Data model

Four classes cover every field the screens show. Fields that only some update types use are nullable. One `TimetableSlot` list drives both today's classes and the free-room finder.

Class	Fields
`AppUser`	`String id, name, email`; `UserRole role` (student, professor); `List<String> courseIds`
`Course`	`String id, code, title, professorName`

Class	Fields
`CourseUpdate`	`String id, courseId, message`; `UpdateType type` (roomChange, cancellation, deadline, note); `DateTime postedAt`; `DateTime? affectedDate, dueAt`; `String? newRoom, deadlineTitle`; `int seenCount`
`TimetableSlot`	`String courseId, room, building`; `int weekday, startMinute, endMinute`; method `bool isBusyAt(DateTime t)`

6. Out of scope

These parts need tools beyond Weeks 1 to 7, so the midterm version runs on sample courses, updates and a timetable stored in Dart lists:

- A backend and database, so a professor's post reaches students' phones, and the real "seen by" count.
- Hero Study and EduPage integration for real courses, logins and timetables.
- Push notifications for urgent changes.

7. Who did what

Member	Student ID	Contribution
Mardon Hazratov (team lead)	240495	Problem research, evidence, document editing
Umarov Istamjon	240460	User flows and wireframes for screens 1 to 6
Usmonov Muhammad Asadxon	240453	Data model
Xushvaqtov Elshod	240444	Presentation slides
Erkinov Sardor	260571	Flutter plan; Room schedule screen (wireframe and plan)

Appendix: evidence notes

Our evidence is our own documented experience as NewUU students, which the requirements accept. The notes below record what we observed.

Source	Date	What we recorded
Course Telegram chat	Oct 13	06:17: professor posts a lecture postponement and asks students to "notify other students who have my class as well." 07:47: the message is forwarded again. 08:36: a student replies "Олдинрок йозмисизми?" ("Couldn't you have written earlier?").
Hero Study app, home screen	Oct 2026	The schedule shows times and rooms, with no way to mark a room change or cancellation. Course content added by professors cannot be opened from the app.
NewUU library, main study area	Oct 2026	During the day nearly every seat is taken, so students look for empty classrooms instead.
EduPage timetable	Oct 2026	Room schedules are published per room and class; there is no view of which rooms are free now.